



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 02

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Program:

```
def binary_search(arr, low, high, key):  
    if low <= high:  
        mid = (low + high) // 2  
        if arr[mid] == key:  
            return mid  
        elif arr[mid] < key:  
            return binary_search(arr, mid + 1, high, key)  
        else:  
            return binary_search(arr, low, mid - 1, key)  
    return -1  
  
arr = [5, 10, 15, 20, 25]  
key = 20  
result = binary_search(arr, 0, len(arr) - 1, key)  
if result != -1:  
    print("Key found at index", result)  
else:  
    print("Key not found")
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def binary_search(arr, low, high, key): 2 if low <= high: 3 mid = (low + high) // 2 4 5 if arr[mid] == key: 6 return mid 7 elif arr[mid] < key: 8 return binary_search(arr, mid + 1, high, key) 9 else: 10 return binary_search(arr, low, mid - 1, key) 11 12 return -1 13 14 15 arr = [5, 10, 15, 20, 25] 16 key = 20 17 18 result = binary_search(arr, 0, len(arr) - 1, key) 19 20 if result != -1: 21 print("Key found at index", result) 22 else: 23 print("Key not found")</pre>			Key found at index 3